

Ruichen Xu

ruichen.xu@stonybrook.edu | Homepage | LinkedIn | GitHub | Google Scholar | OpenReview

SUMMARY

Ph.D. candidate in Computational Applied Mathematics building **Python/PyTorch models and reproducible training and evaluation pipelines**. Research spans sparse-observation inference, scientific representation learning, generative models and optimization, with a **560,000-record benchmark** and coauthored ICLR, TMLR and IJCNN papers. Expected graduation: 2027.

EDUCATION

Stony Brook University	Aug 2022 – Present
Ph.D., Computational Applied Mathematics GPA: 4.00/4.00 Expected 2027	Stony Brook, NY
New York University – Courant Institute	Sep 2020 – May 2022
M.S., Mathematics GPA: 4.00/4.00	New York, NY
University of California, Davis	Sep 2019 – Jun 2020
M.S., Statistics GPA: 3.94/4.00	Davis, CA
Beijing University of Chemical Technology	Sep 2015 – Jun 2019
Bachelor's degree, Financial Mathematics Minor: Commercial Management	Beijing, China

TECHNICAL SKILLS

Programming & Computing: Python, PyTorch; GPU/HPC experiment workflows, SLURM.

Modeling: Representation learning, neural operators, graph learning, diffusion models, reinforcement learning.

Evaluation & Optimization: Benchmarking, ablations, distribution-shift evaluation, simulated annealing, molecular search.

RESEARCH EXPERIENCE

Stony Brook University Doctoral Research	Aug 2022 – Present
Advisor: Prof. Yuefan Deng	

PDE-OBS: Controlled Evaluation Across Observation Patterns	2025 – Present
---	----------------

- Built a **560,000-record resource** of fields and trajectories across **seven PDE families**.
- Adapted **seven baselines** to mask-aware inputs with fixed physical records, prediction targets and data splits.
- Evaluated **441 trained configurations / 3,969 pattern tests** on a **14,000-record study subset**.
- Found higher mean cross-pattern error in all **49 PDE-method pairs** versus matched-pattern evaluation.
- Tested mixed-pattern training on five completed pairs; released reproducible observation-shift scripts. [Code]

JEPA for Science	Jan 2026 – Present
-------------------------	--------------------

- **HP-JEPA:** Developed multi-resolution joint-embedding prediction for scientific graph classification and regression; outperformed Graph-JEPA on **6 of 8 evaluated tasks**.
- **JENO:** Built latent pretraining for sparse inverse PDE recovery, followed by supervised encoder/readout fine-tuning; used complete fields during training and only partial observations at inference.
- **SCOPE:** Coupled latent prediction with a shared physical decoder; evaluated **10 forward/inverse tasks** with deterministic one-pass inference and decoder-only adaptation.

ORACLE: LLM-Guided Molecular Optimization	2025 – Present
--	----------------

- Built a **simulated-annealing pipeline** linking **LLM-proposed molecular edits**, validity checks, candidate scoring and acceptance decisions for structure-based molecular optimization.
- Implemented configurable docking, QED, synthetic-accessibility and diversity objectives to compare candidate molecules and expose multi-objective trade-offs.
- Tracked search configurations and ablations across edit operators, cooling schedules and acceptance rules; compared optimization curves, molecular validity and Pareto trade-offs.

Physics-Aware Dynamics, Diffusion & Optimization	Selected 2025 Work
---	--------------------

- **KAR-HNN / position-only dynamics:** Co-led Kolmogorov-Arnold Hamiltonian learning (**IJCNN; co-first author, project lead**) and led velocity-inferred modeling from position measurements (**NYSDS; first author**).
- **APOD / RL-QESA:** Developed observation- and PDE-guided diffusion sampling and reinforcement-learned annealing schedules; **two first-author ICML 2025 workshop papers**.

Efficient Generative Models & VLM Post-Training	Selected 2025–2026 Work
--	-------------------------

- **GAGA (ICLR 2026):** Coauthored a Gaussianity-aware approximation for 3D molecular diffusion; the study reports **up to 40% shorter generative trajectories** on evaluated configurations.
- **VLM distillation (preprint):** Coauthored student-curriculum coupling for temporal video grounding; the study reports **+5.1% relative mean recall** versus Video-OPD, **60.0% fewer examples** and **50.4% less training time**.

SELECTED PUBLICATIONS & MANUSCRIPTS

Name in bold; * denotes co-first author. Full list: [Homepage](#) / [Google Scholar](#)

Conference & Journal Papers

- [1] **GAGA: Gaussianity-Aware Gaussian Approximation for Efficient 3D Molecular Generation.** J. Qu, W. Gao, **Ruichen Xu**, Y. Liu. *ICLR 2026*.
- [2] **Discretization-invariance? On the Discretization Mismatch Errors in Neural Operators.** W. Gao, **Ruichen Xu**, Y. Deng, Y. Liu. *ICLR 2025*.
- [3] **Kolmogorov-Arnold Representation for Symplectic Learning: Advancing Hamiltonian Neural Networks.** Z. Wu, **Ruichen Xu***, et al. *IJCNN 2025*. Co-first author; project lead.
- [4] **Dynamic Schwartz-Fourier Neural Operator for Enhanced Expressive Power.** W. Gao, J. Luo, **Ruichen Xu**, Y. Liu. *TMLR 2025*.
- [5] **Coordinate Transform Fourier Neural Operators for Symmetries in Physical Modeling.** W. Gao, **Ruichen Xu**, H. Wang, Y. Liu. *TMLR 2024*.
- [6] **Towards Principled Test-Time Adaptation for Time Series Forecasting.** H. Wang, **Ruichen Xu**, et al. *TMLR 2026*; *accepted with minor revision*.
- [7] **Multistep Generative Backmapping of Coarse-Grained Structures.** G. Kementzidis, E. Wong, J. Nicholson, **Ruichen Xu**, Y. Deng. *Computer Physics Communications 327, 110286 (2026)*.
- [8] **Velocity-Inferred Hamiltonian Networks: Symplectic Dynamics from Position-Only Observations.** **Ruichen Xu**, C. Yu, et al. *NYSDS 2025, SIAM Proceedings*.
- [9] **Boundary-Informed Method of Lines for Physics-Informed Neural Networks.** M. Cederholm, S. Wang, H. Wang, **Ruichen Xu**, Y. Deng. *NYSDS 2025, SIAM Proceedings*. Co-corresponding author.
- [10] **Physics-Informed Active Learning via Functional Simulated Annealing for Neural Operator.** A. Ding, S. Wang, H. Wang, **Ruichen Xu**, Y. Deng. *NYSDS 2025, SIAM Proceedings*. Co-corresponding author.

Workshop Papers

- [11] **APOD: Adaptive PDE-Observation Diffusion for Physics-Constrained Sampling.** **Ruichen Xu**, H. Wang, et al. *ICML 2025 Workshop on Assessing World Models (poster)*.
- [12] **RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing.** **Ruichen Xu**, K. Li, et al. *2nd AI for Math Workshop at ICML 2025 (poster)*.

Preprints & Manuscripts

- [13] **SCOPE: Observation-Conditioned Full-Target Prediction for Sparse PDE Inference.** **Ruichen Xu**, et al. *arXiv:2609.36527, 2026*; *under review at ICLR 2027*.
- [14] **PDE-OBS: Controlled Evaluation Across Observation Patterns.** **Ruichen Xu**, et al. *arXiv:2609.36521, 2026*.
- [15] **HP-JEPA: Hierarchical Partitioning for Multi-Resolution Graph Joint-Embedding Predictive Learning.** **Ruichen Xu**, et al. *arXiv:2608.00491, 2026*.
- [16] **JENO: Full-Field Latent Prediction for Sparse Inverse PDE Inference.** **Ruichen Xu**, et al. *Manuscript, 2026*.
- [17] **Less Data, Better Timing: Student-Curriculum Coupling for VLM On-Policy Distillation in Temporal Video Grounding.** J. Qiu, Y. Kim, **Ruichen Xu**, et al. *arXiv:2609.40055, 2026*.
- [18] **The Impact of Move Schemes on Simulated Annealing Performance.** **Ruichen Xu**, H. Wang, Y. Deng. *arXiv:2504.17949, 2025*.

MENTORING & TEACHING

AI3 REU, Stony Brook University | Research Mentor

Jun – Jul 2026

Mentored AI4Science project scoping, experiment design and implementation; guided reproducible workflows and research presentations.

Stony Brook University | Instructor & Teaching Assistant

Aug 2022 – Present

Instructor: Statistical Laboratory, Numerical Analysis and Differential Equations. Designed lectures and assessments; led recitations, office hours and grading as a teaching assistant.

NYU Courant | Recitation Leader & Grader

2020 – 2022

Led recitations in Mathematics for Economics II and Probability, Statistics & Decision Making; graded Analysis and Special Topics coursework.

RECOGNITION, ACADEMIC SERVICE & TALKS

ICML 2026 Gold Reviewer. Conference service: ICLR (2026/2027), NeurIPS (2026), IJCNN (2025/2026), KDD (2026/2027); AAAI 2027 Program Committee.

Journal / workshop reviewing: TMLR, IEEE TNNLS, Neurocomputing; AI for Math at ICML 2025.

IACS presentations: AI for PDEs (2025); Diffusion4PDE (2025); Active Learning for Neural Operators (2024).